



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design) Practice 3: Projective Geometry and Camera Model

Spring 2026

Instructor: Hao Li

Outline

- Wonder of Projective Geometry
- Pinhole Camera Model
- Use of Homography

《墨子·经说下》

景，光之人煦若射，
下者之人也高，高者之人也下。
足敝下光，故成景于上；
首敝上光，故成景于下。
在远近有端，与于光，
故景庠（障）内也。

景到，在乎有端，与景长，
说在端

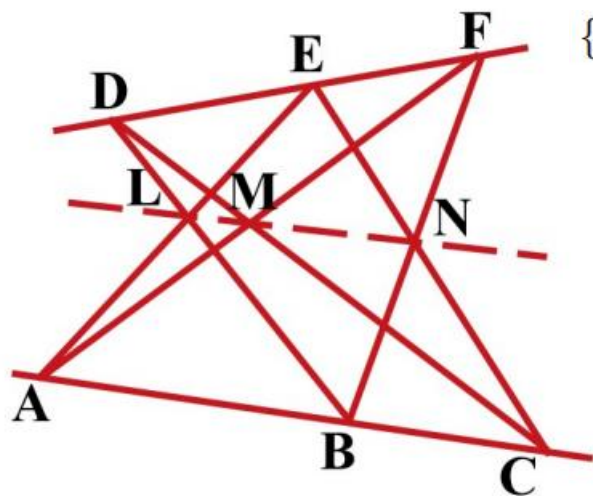


THREE-DIMENSIONAL COMPUTER VISION



Perspective painting technique

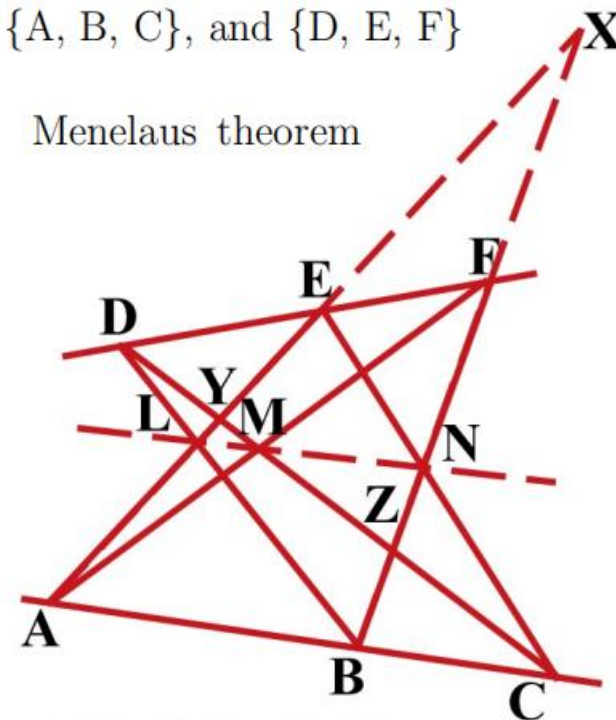
Pappus Theorem



$\{A, M, F\}$, $\{B, L, D\}$, $\{C, N, E\}$, $\{A, B, C\}$, and $\{D, E, F\}$

$$\begin{aligned} \frac{|ZM|}{|MY|} \cdot \frac{|YA|}{|AX|} \cdot \frac{|XF|}{|FZ|} &= 1, \\ \frac{|YL|}{|LX|} \cdot \frac{|XB|}{|BZ|} \cdot \frac{|ZD|}{|DY|} &= 1, \\ \frac{|XN|}{|NZ|} \cdot \frac{|ZC|}{|CY|} \cdot \frac{|YE|}{|EX|} &= 1, \\ \frac{|YC|}{|CZ|} \cdot \frac{|ZB|}{|BX|} \cdot \frac{|XA|}{|AY|} &= 1, \\ \frac{|XE|}{|EY|} \cdot \frac{|YD|}{|DZ|} \cdot \frac{|ZF|}{|FX|} &= 1. \end{aligned}$$

Menelaus theorem



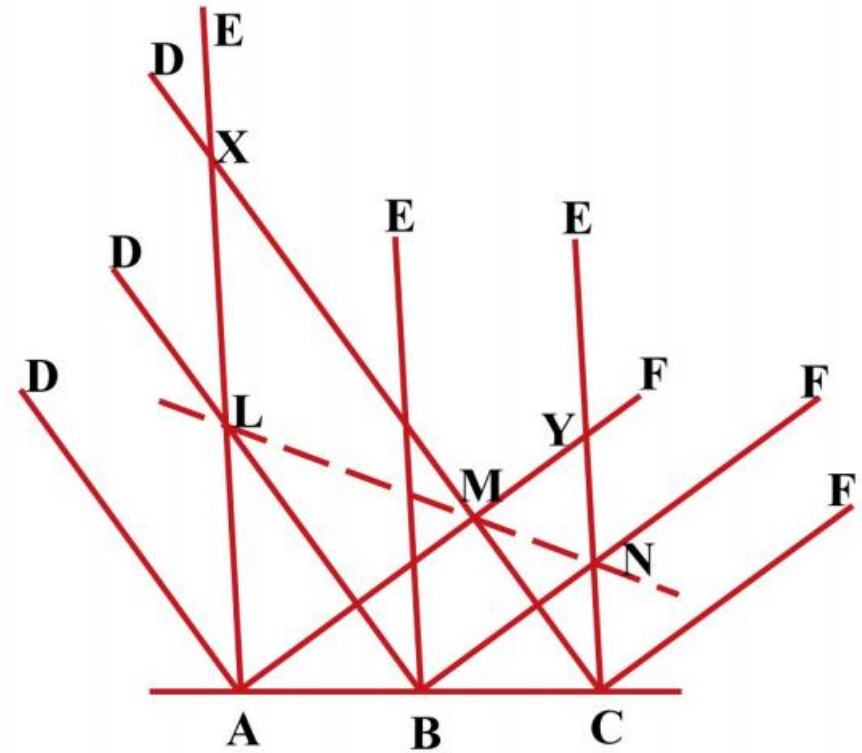
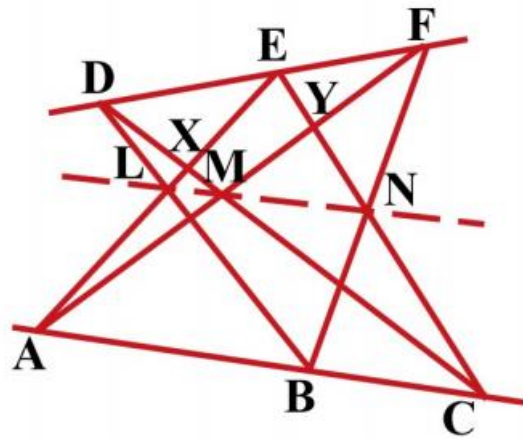
$$\begin{aligned} \prod \text{left sides} &= \frac{|ZM|}{|MY|} \cdot \frac{|YA|}{|AX|} \cdot \frac{|XF|}{|FZ|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XB|}{|BZ|} \cdot \frac{|ZD|}{|DY|} \cdot \frac{|XN|}{|NZ|} \cdot \frac{|ZC|}{|CY|} \cdot \frac{|YE|}{|EX|} \\ &\quad \cdot \frac{|YC|}{|CZ|} \cdot \frac{|ZB|}{|BX|} \cdot \frac{|XA|}{|AY|} \cdot \frac{|XE|}{|EY|} \cdot \frac{|YD|}{|DZ|} \cdot \frac{|ZF|}{|FX|} \\ &= \frac{|ZM|}{|MY|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XN|}{|NZ|} = 1. \end{aligned}$$

Invariance In Projective Geometry

- *Invariance 1*: A point after perspective mapping (or perspective transform) is still a point.
- *Invariance 2*: A line after perspective mapping is still a line.
- *Invariance 3*: Coinciding lines after perspective mapping still coincide.
- *Invariance 4*: Colinear points after perspective mapping are still colinear.



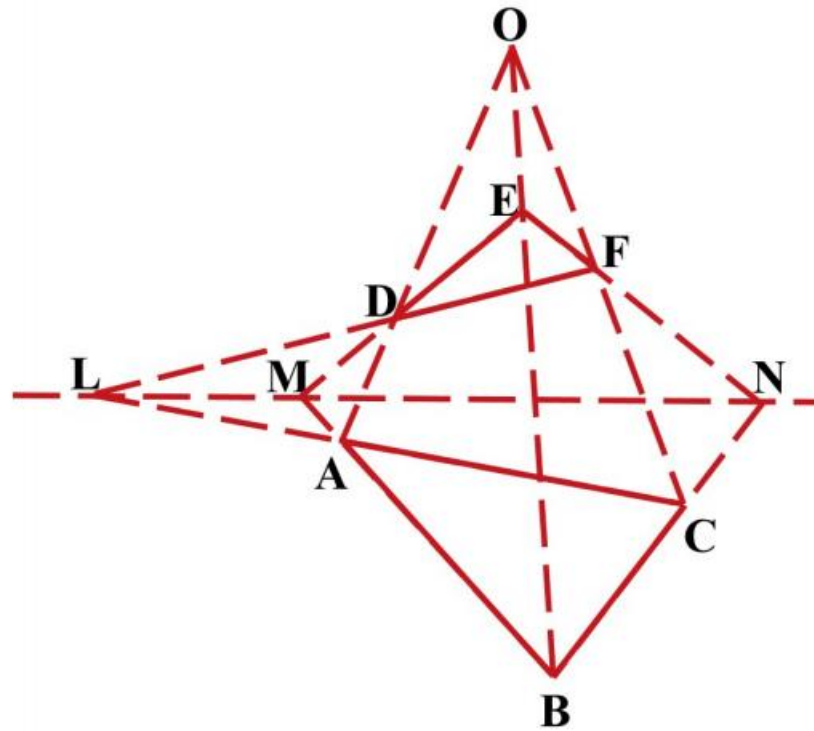
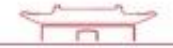
Pappus Theorem - PG Perspective



$$\frac{|XL|}{|LA|} = \frac{|CB|}{|BA|} = \frac{|CN|}{|NY|}$$

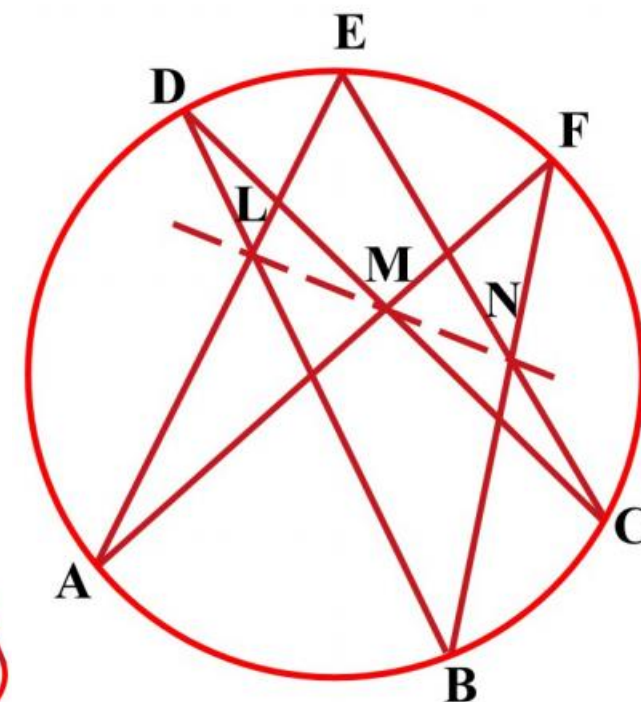
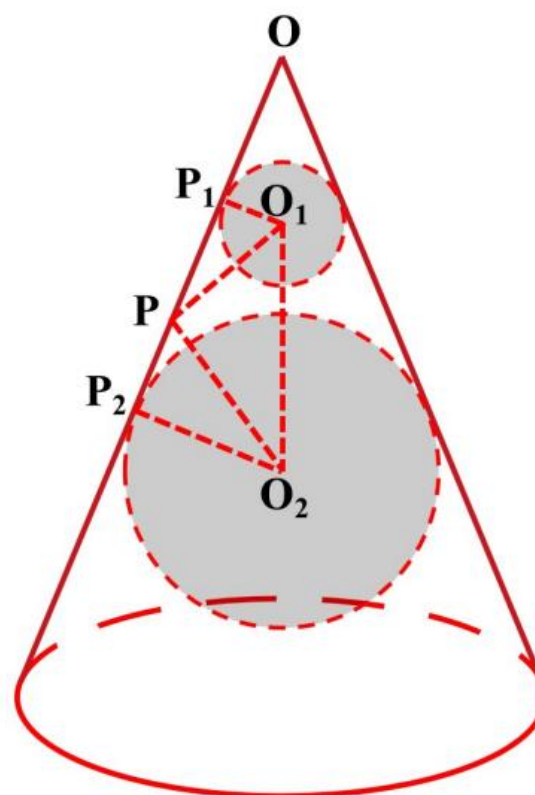
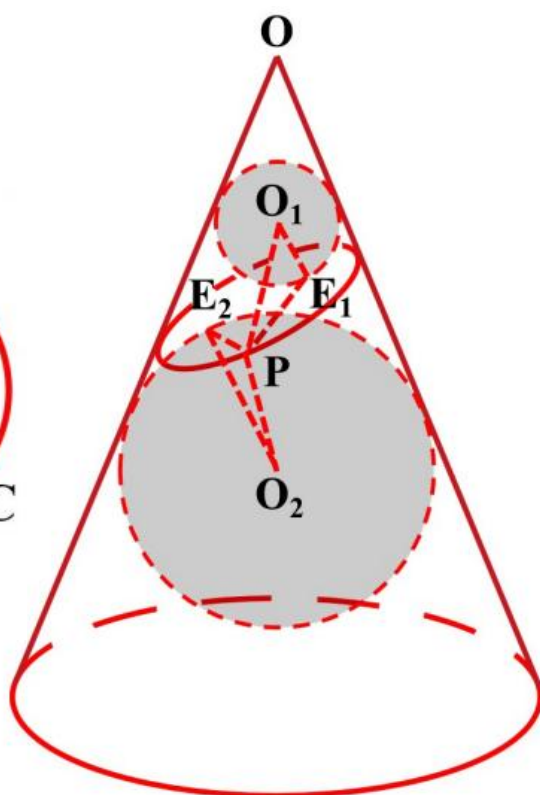
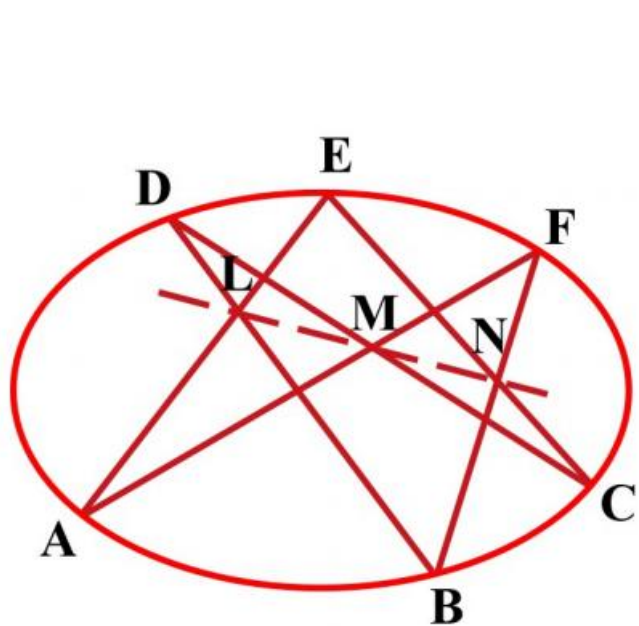


Desargue Theorem - PG Perspective



of a tetrahedron intersected by two planes in the three-dimensional space. The tetrahedron is formed by the four triangles O - A - B , O - B - C , O - C - A , and A - B - C . The two planes are spanned by the triangle A - B - C and the triangle D - E - F respectively.

Pascal Theorem - PG Perspective

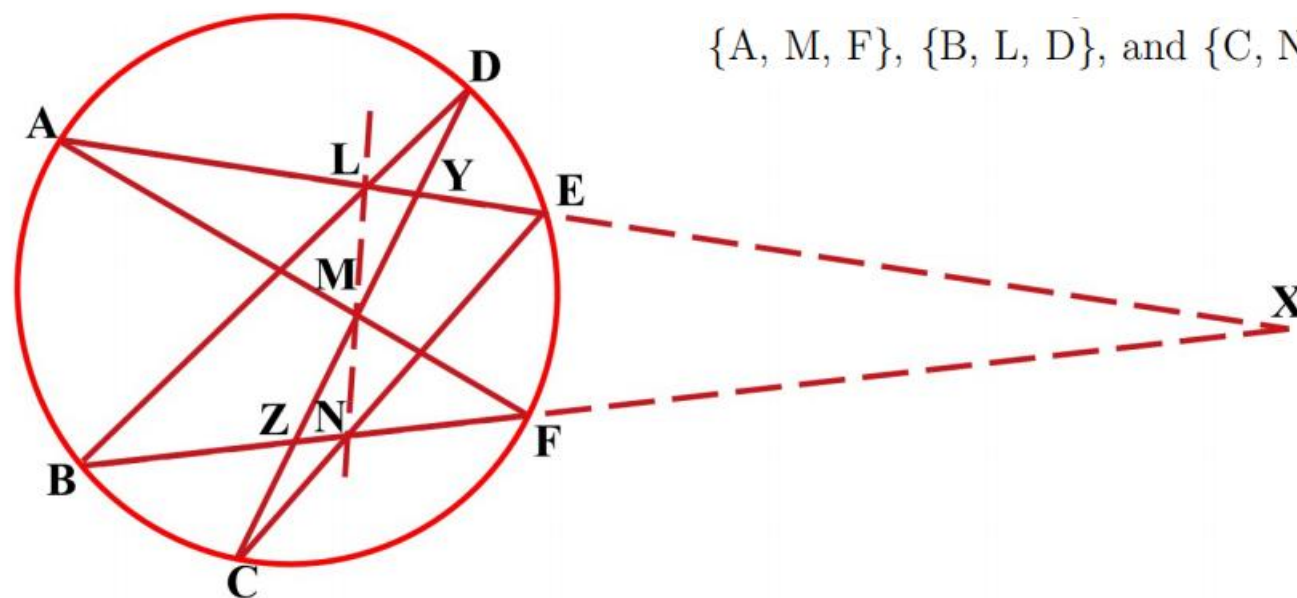


$$|E_1P| = \sqrt{|O_1P|^2 - |O_1E_1|^2} = \sqrt{|O_1P|^2 - |O_1P_1|^2} = |P_1P|$$

$$|E_2P| = \sqrt{|O_2P|^2 - |O_2E_2|^2} = \sqrt{|O_2P|^2 - |O_2P_2|^2} = |P_2P|$$

$$|E_1P| + |E_2P| = |P_1P| + |P_2P| = |P_1P_2| \quad (\text{a constant})$$

Pascal Theorem - PG Perspective



$\{A, M, F\}$, $\{B, L, D\}$, and $\{C, N, E\}$

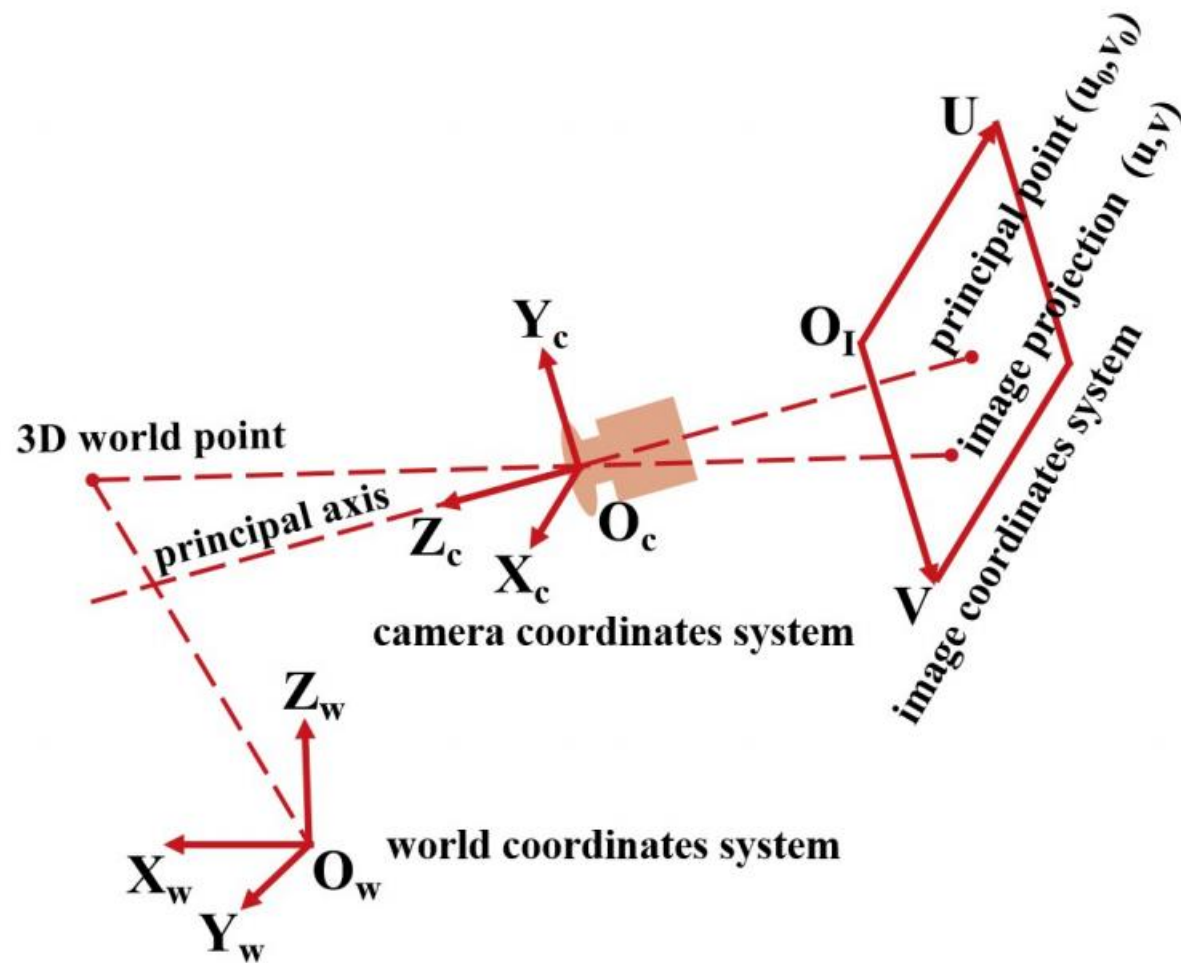
$$\begin{aligned}
 \frac{|ZM|}{|MY|} \cdot \frac{|YA|}{|AX|} \cdot \frac{|XF|}{|FZ|} &= 1 & \prod \text{left sides} &= \frac{|ZM|}{|MY|} \cdot \frac{|YA|}{|AX|} \cdot \frac{|XF|}{|FZ|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XB|}{|BZ|} \cdot \frac{|ZD|}{|DY|} \cdot \frac{|XN|}{|NZ|} \cdot \frac{|ZC|}{|CY|} \cdot \frac{|YE|}{|EX|} \\
 \frac{|YL|}{|LX|} \cdot \frac{|XB|}{|BZ|} \cdot \frac{|ZD|}{|DY|} &= 1 & &= \frac{|ZM|}{|MY|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XN|}{|NZ|} \cdot \frac{|XF| \cdot |XB|}{|AX| \cdot |EX|} \cdot \frac{|YA| \cdot |YE|}{|CY| \cdot |DY|} \cdot \frac{|ZD| \cdot |ZC|}{|BZ| \cdot |FZ|} \\
 \frac{|XN|}{|NZ|} \cdot \frac{|ZC|}{|CY|} \cdot \frac{|YE|}{|EX|} &= 1 & &= \frac{|ZM|}{|MY|} \cdot \frac{|YL|}{|LX|} \cdot \frac{|XN|}{|NZ|} = 1,
 \end{aligned}$$



DISCUSSION

Q & A

World, Camera, Image Coordinates Systems



world coordinates system
(WCS)

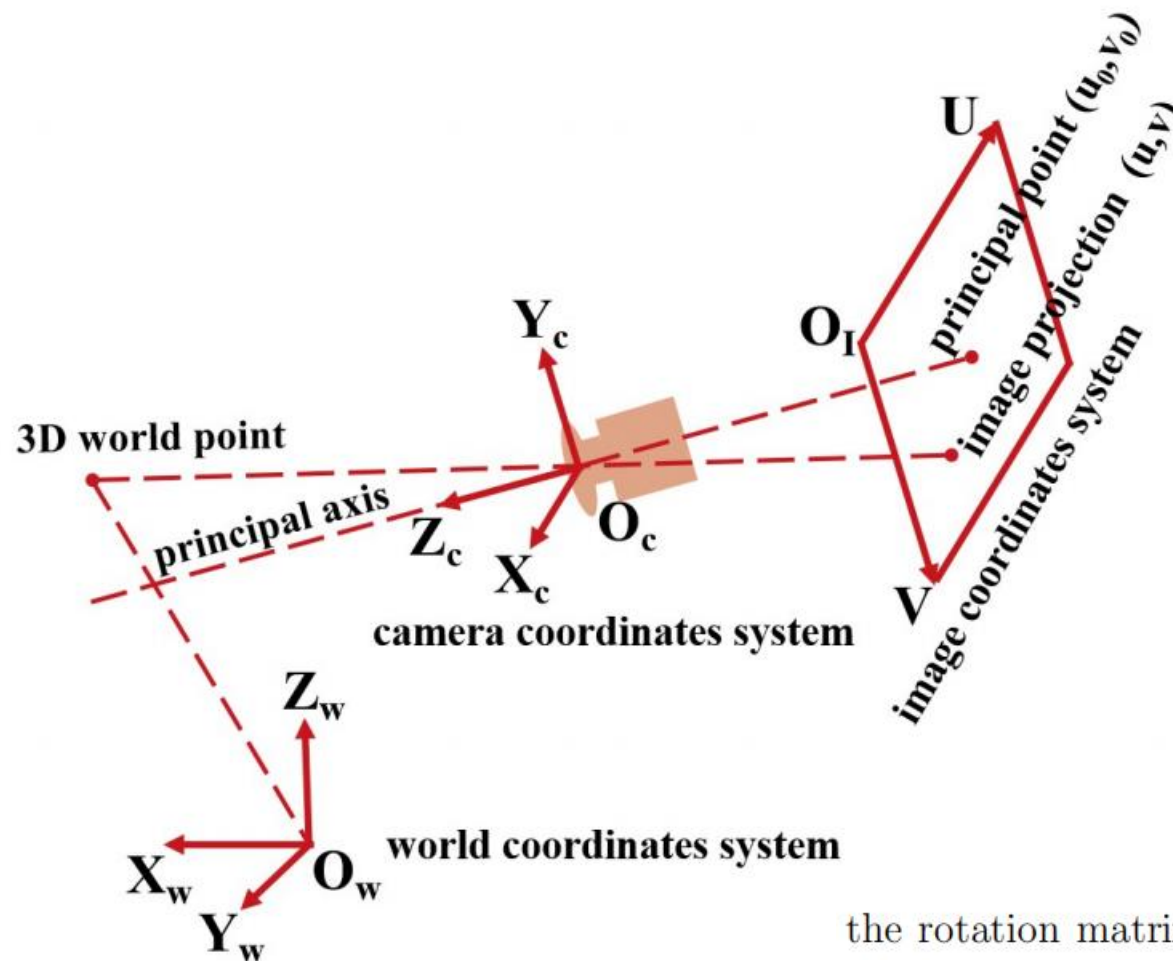


camera coordinates system
(CCS)



image coordinates system
(ICS)

World CS to Camera CS



world coordinates system
(WCS)



camera coordinates system
(CCS)

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = R_{wc} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + T_{wc}$$

$$P_c = R_{wc}P_w + T_{wc}$$

the rotation matrix R_{wc} and the translation vector T_{wc}

camera extrinsic parameters

Basic 3D Rotations



$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$\mathbf{R}_x(\theta) = \begin{bmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix}$$

Properties of the Rotation Matrix

Orthonormality

$$\mathbf{R}^T \mathbf{R} = \mathbf{R} \mathbf{R}^T = \mathbf{I}$$

$$\mathbf{R} \equiv \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\begin{bmatrix} r_{1j} & r_{2j} & r_{3j} \end{bmatrix} \begin{bmatrix} r_{1j} \\ r_{2j} \\ r_{3j} \end{bmatrix} = r_{1j}^2 + r_{2j}^2 + r_{3j}^2 = 1, \quad j \in \{1,2,3\}$$

$$\begin{bmatrix} r_{i1} & r_{i2} & r_{i3} \end{bmatrix} \begin{bmatrix} r_{i1} \\ r_{i2} \\ r_{i3} \end{bmatrix} = r_{i1}^2 + r_{i2}^2 + r_{i3}^2 = 1, \quad i \in \{1,2,3\}$$

$$\begin{bmatrix} r_{1j} & r_{2j} & r_{3j} \end{bmatrix} \begin{bmatrix} r_{1k} \\ r_{2k} \\ r_{3k} \end{bmatrix} = r_{1j}r_{1k} + r_{2j}r_{2k} + r_{3j}r_{3k} = 0, \quad j \neq k$$

$$\begin{bmatrix} r_{i1} & r_{i2} & r_{i3} \end{bmatrix} \begin{bmatrix} r_{k1} \\ r_{k2} \\ r_{k3} \end{bmatrix} = r_{i1}r_{k1} + r_{i2}r_{k2} + r_{i3}r_{k3} = 0, \quad i \neq k$$

$$\mathbf{R}^{-1} = \mathbf{R}^T = \begin{bmatrix} r_{11} & r_{21} & r_{31} \\ r_{12} & r_{22} & r_{32} \\ r_{13} & r_{23} & r_{33} \end{bmatrix}$$

Properties of the Rotation Matrix

Orthonormality

$$\mathbf{R}^T \mathbf{R} = \mathbf{R} \mathbf{R}^T = \mathbf{I}$$

$$\mathbf{R} \equiv \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

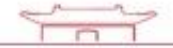
How many degrees of freedom (DoF) are there in the rotation matrix ?

$$r_{1j}^2 + r_{2j}^2 + r_{3j}^2 = 1, \quad j \in \{1, 2, 3\}$$

$$r_{1j}r_{1k} + r_{2j}r_{2k} + r_{3j}r_{3k} = 0, \quad j \neq k$$

$$\text{DoF} = 9 - 6 = 3$$

World CS to Camera CS



$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \mathbf{R}_{wc} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \mathbf{T}_{wc}$$

If $\mathbf{T}_{wc} = \mathbf{0}$ then $\frac{x_c^2 + y_c^2 + z_c^2}{x_w^2 + y_w^2 + z_w^2} = ?$

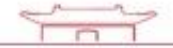
world coordinates system
(WCS)



camera coordinates system
(CCS)

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \mathbf{0}; \mathbf{R}_{wc} = \begin{bmatrix} \sqrt{0.5} & -\sqrt{0.5} & 0 \\ \sqrt{0.5} & \sqrt{0.5} & 0 \\ 0 & 0 & 1 \end{bmatrix}; \mathbf{T}_{wc} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = ?$$

3D CS-to-CS Transformation Concatenation



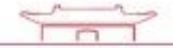
$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \mathbf{R}_{12} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \mathbf{T}_{12}$$

$$\begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \mathbf{R}_{23} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} + \mathbf{T}_{23}$$

$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{12} & \mathbf{T}_{12} \\ & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_3 \\ y_3 \\ z_3 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{23} & \mathbf{T}_{23} \\ & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{23} & \mathbf{T}_{23} \\ & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_{12} & \mathbf{T}_{12} \\ & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix}$$

3D CS-to-CS Transformation Inverse



$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \mathbf{R}_{12} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \mathbf{T}_{12}$$

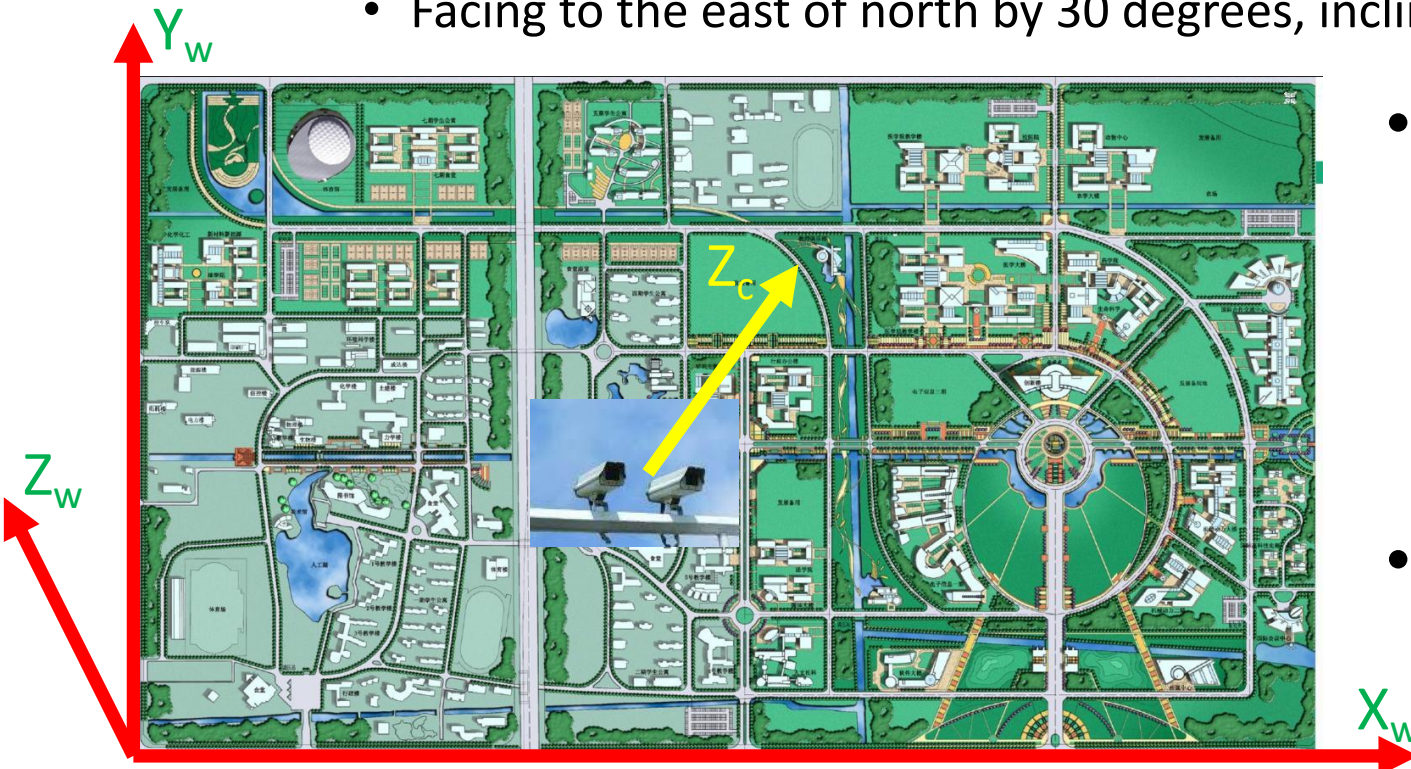
$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{12} & \mathbf{T}_{12} \\ & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{12} & \mathbf{T}_{12} \\ & 1 \end{bmatrix}^{-1} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{12}^T & -\mathbf{R}_{12}^T \mathbf{T}_{12} \\ & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ 1 \end{bmatrix}$$

Comprehensive Example

- Street-View Camera

- “MapForUI.jpg” : 803x1404 pixels, each pixel representing 1.7 meters in the WCS
- Camera configuration
 - Located at (800, 600) meters, at a height of 10 meters
 - Facing to the east of north by 30 degrees, inclining downward by 15 degrees



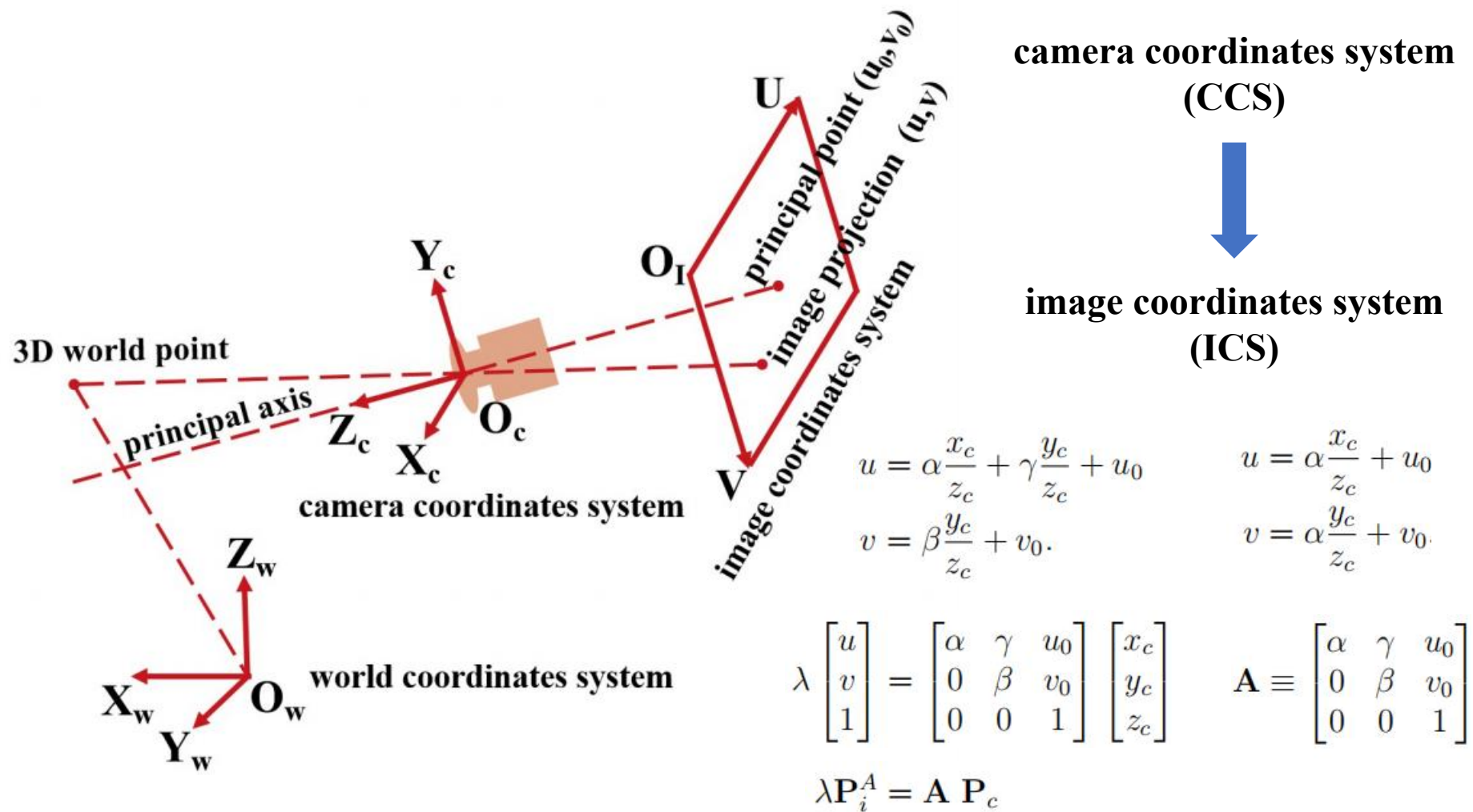
- Intermediate-Pose Approach

- Removal of inclination
- Making the Z-axis upward
- Alignment with the WCS
- Final translation

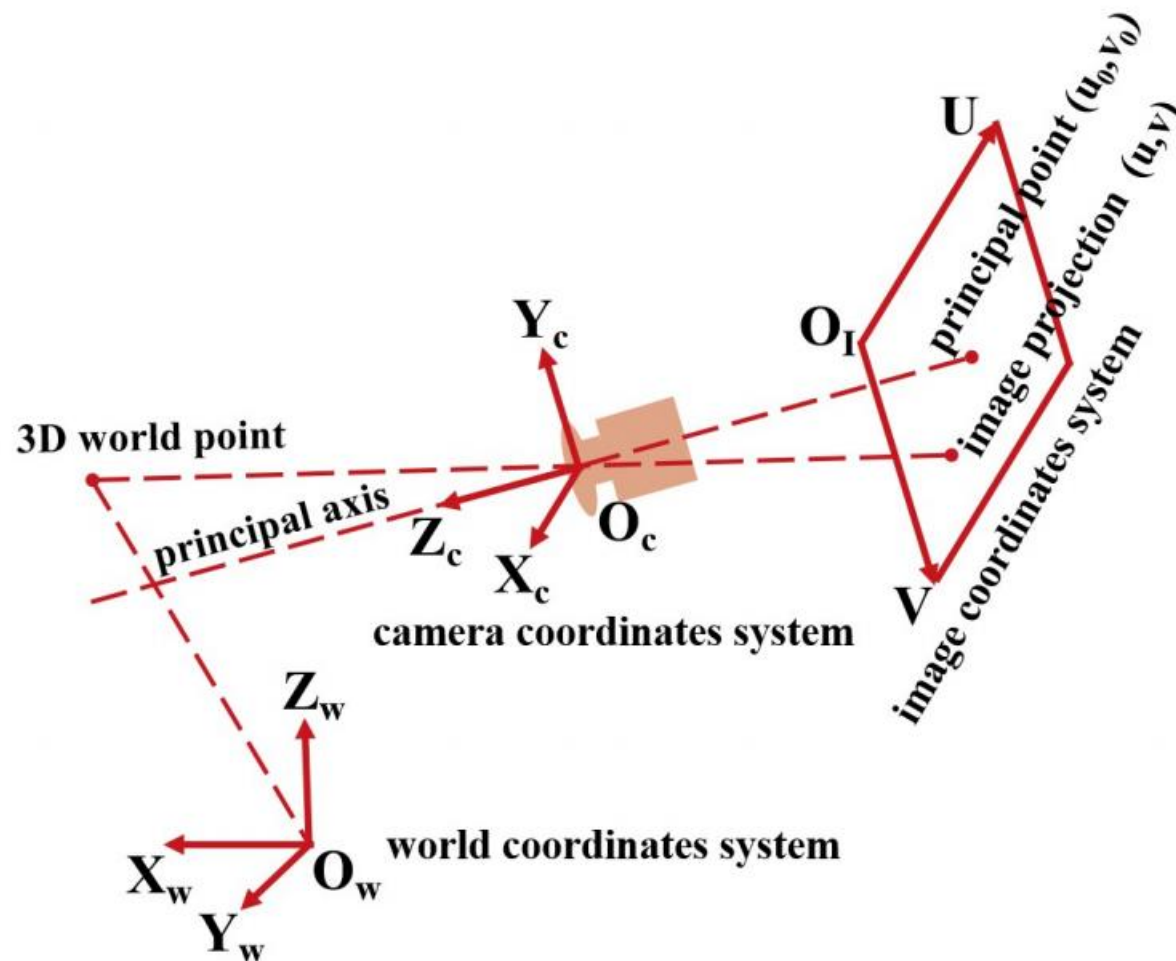
- Base-Vector Approach

- Project each base vector

Camera CS to Image CS



Camera CS to Image CS



camera coordinates system
(CCS)



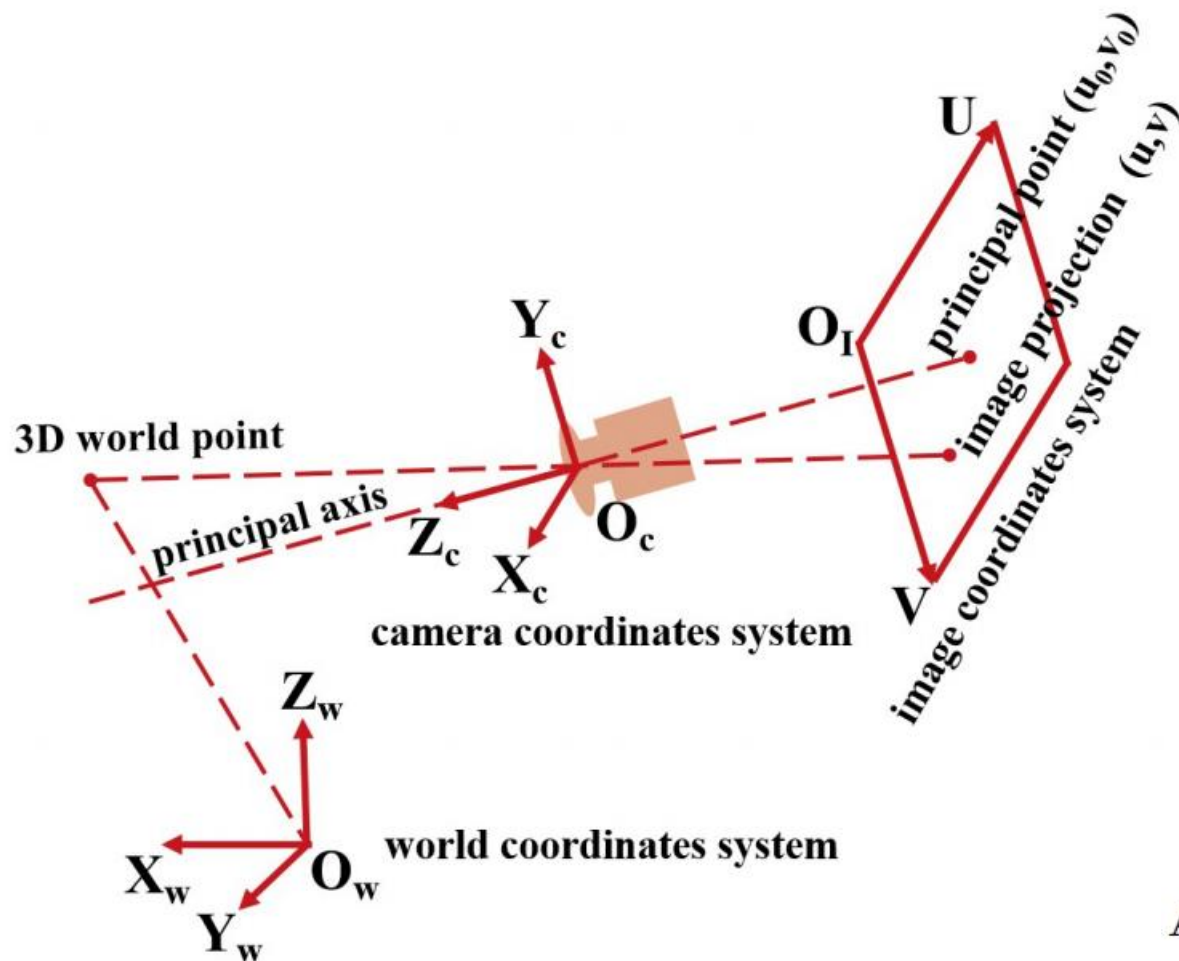
image coordinates system
(ICS)

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} \mathbf{P}_c$$

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \lambda \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

Camera CS to Image CS



camera coordinates system
(CCS)



image coordinates system
(ICS)

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} \mathbf{P}_c$$

$\mathbf{A}$

camera intrinsic matrix

$\alpha, \beta, \gamma, u_0,$ and v_0

camera intrinsic parameters

Pay attention to signs in practice

Perspective Mapping

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \mathbf{R}_{wc} \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + \mathbf{T}_{wc}$$

$$\mathbf{P}_c = \mathbf{R}_{wc} \mathbf{P}_w + \mathbf{T}_{wc}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} \mathbf{P}_c$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \mathbf{P}_w^A$$

$$\begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$\mathbf{P}_c = [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \mathbf{P}_w^A$$

**world coordinates system
(WCS)**



**image coordinates system
(ICS)**

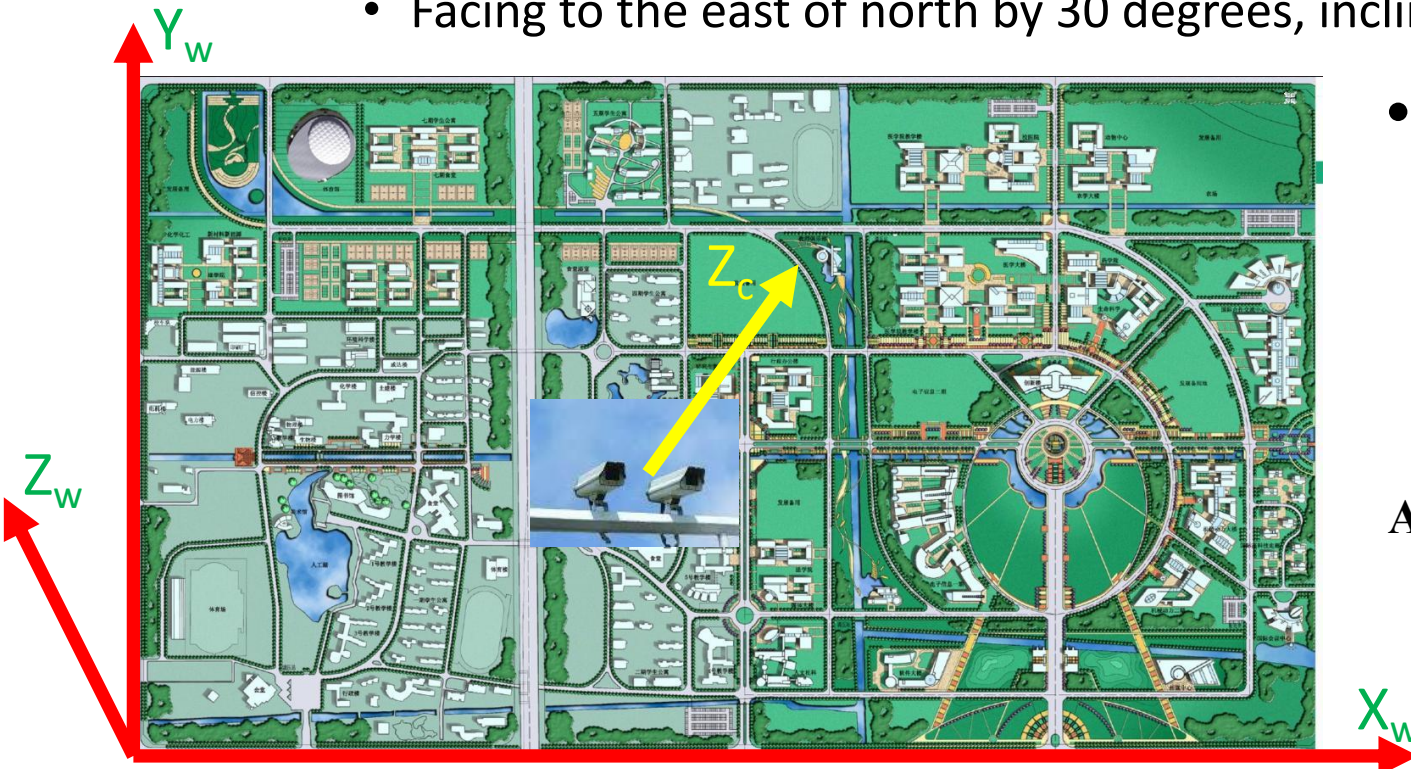
$$\mathbf{M} \equiv \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \equiv \mathbf{A} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}]$$

perspective mapping matrix

Comprehensive Example - Revisited

- Street-View Camera

- “MapForUI.jpg” : 803x1404 pixels, each pixel representing 1.7 meters in the WCS
- Camera configuration
 - Located at (800, 600) meters, at a height of 10 meters
 - Facing to the east of north by 30 degrees, inclining downward by 15 degrees



- World to Image

- Compute image projections of P_1, P_2

$$\mathbf{A} = \begin{bmatrix} -300 & 0 & 160 \\ 0 & -300 & 120 \\ 0 & 0 & 1 \end{bmatrix}$$

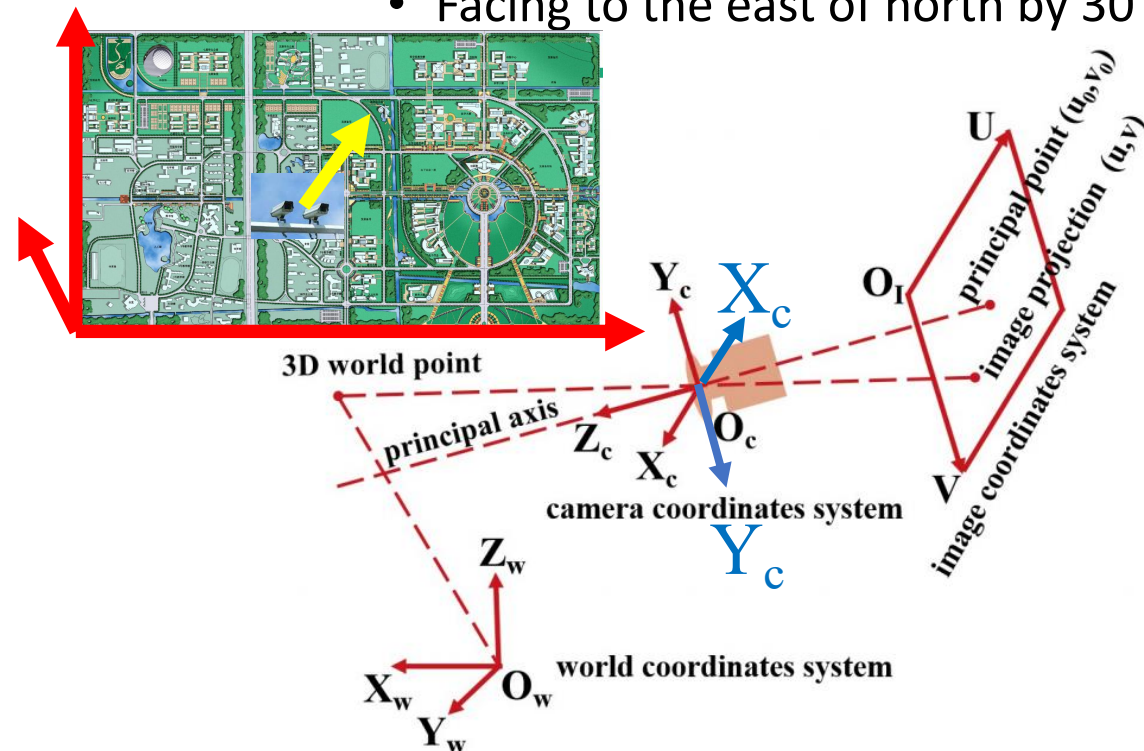
$$\mathbf{P}_1 = \begin{bmatrix} 825 \\ 643 \\ 0 \end{bmatrix} \quad \mathbf{P}_2 = \begin{bmatrix} 850 \\ 687 \\ 0 \end{bmatrix}$$

$$\mathbf{P}_3 = \begin{bmatrix} 825 \\ 625 \\ 0 \end{bmatrix} \quad \mathbf{P}_4 = \begin{bmatrix} 850 \\ 650 \\ 0 \end{bmatrix}$$

Comprehensive Example - Revisited

- Street-View Camera

- “MapForUI.jpg” : 803x1404 pixels, each pixel representing 1.7 meters in the WCS
- Camera configuration
 - Located at (800, 600) meters, at a height of 10 meters
 - Facing to the east of north by 30 degrees, inclining downward by 15 degrees



- Positive Camera Intrinsic Matrix

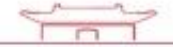
- How should the CCS be pose?

$$\mathbf{A}_+ = \begin{bmatrix} 300 & 0 & 160 \\ 0 & 300 & 120 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{P}_1 = \begin{bmatrix} 825 \\ 643 \\ 0 \end{bmatrix} \quad \mathbf{P}_2 = \begin{bmatrix} 850 \\ 687 \\ 0 \end{bmatrix}$$

$$\mathbf{P}_3 = \begin{bmatrix} 825 \\ 625 \\ 0 \end{bmatrix} \quad \mathbf{P}_4 = \begin{bmatrix} 850 \\ 650 \\ 0 \end{bmatrix}$$

Perspective Mapping



$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}$$

$$\lambda \mathbf{P}_i^A = \mathbf{A} [\mathbf{R}_{wc} \quad \mathbf{T}_{wc}] \mathbf{P}_w^A$$

$$\mathbf{M} \equiv \begin{bmatrix} \mathbf{M}_{1,:} \\ \mathbf{M}_{2,:} \\ \mathbf{M}_{3,:} \end{bmatrix} \equiv \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix}$$

perspective mapping matrix

$$u = \frac{\mathbf{M}_{1,:} \mathbf{P}_w^A}{\mathbf{M}_{3,:} \mathbf{P}_w^A} = \frac{m_{11}x_w + m_{12}y_w + m_{13}z_w + m_{14}}{m_{31}x_w + m_{32}y_w + m_{33}z_w + m_{34}}$$
$$v = \frac{\mathbf{M}_{2,:} \mathbf{P}_w^A}{\mathbf{M}_{3,:} \mathbf{P}_w^A} = \frac{m_{21}x_w + m_{22}y_w + m_{23}z_w + m_{24}}{m_{31}x_w + m_{32}y_w + m_{33}z_w + m_{34}}$$

**world coordinates system
(WCS)**



**image coordinates system
(ICS)**

Revisit Common Sense of Projective Geometry



Vanishing point

$$L_w : \quad \mathbf{P}_w(t) \equiv [x_w^0 + x_w t \quad y_w^0 + y_w t \quad z_w^0 + z_w t]^T$$

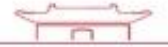
$$u(t) = \frac{\mathbf{M}_{1,:} \mathbf{P}_w^A(t)}{\mathbf{M}_{3,:} \mathbf{P}_w^A(t)} = \frac{(m_{11}x_w^0 + m_{12}y_w^0 + m_{13}z_w^0 + m_{14}) + (m_{11}x_w + m_{12}y_w + m_{13}z_w)t}{(m_{31}x_w^0 + m_{32}y_w^0 + m_{33}z_w^0 + m_{34}) + (m_{31}x_w + m_{32}y_w + m_{33}z_w)t}$$

$$v(t) = \frac{\mathbf{M}_{2,:} \mathbf{P}_w^A(t)}{\mathbf{M}_{3,:} \mathbf{P}_w^A(t)} = \frac{(m_{21}x_w^0 + m_{22}y_w^0 + m_{23}z_w^0 + m_{24}) + (m_{21}x_w + m_{22}y_w + m_{23}z_w)t}{(m_{31}x_w^0 + m_{32}y_w^0 + m_{33}z_w^0 + m_{34}) + (m_{31}x_w + m_{32}y_w + m_{33}z_w)t}$$

As $t \rightarrow \infty$, the $[u(t) \quad v(t)]^T$ of all the parallel lines approach the common limit

$$\begin{cases} u(t) & \rightarrow \frac{m_{11}x_w + m_{12}y_w + m_{13}z_w}{m_{31}x_w + m_{32}y_w + m_{33}z_w} \\ v(t) & \rightarrow \frac{m_{21}x_w + m_{22}y_w + m_{23}z_w}{m_{31}x_w + m_{32}y_w + m_{33}z_w} \end{cases}$$

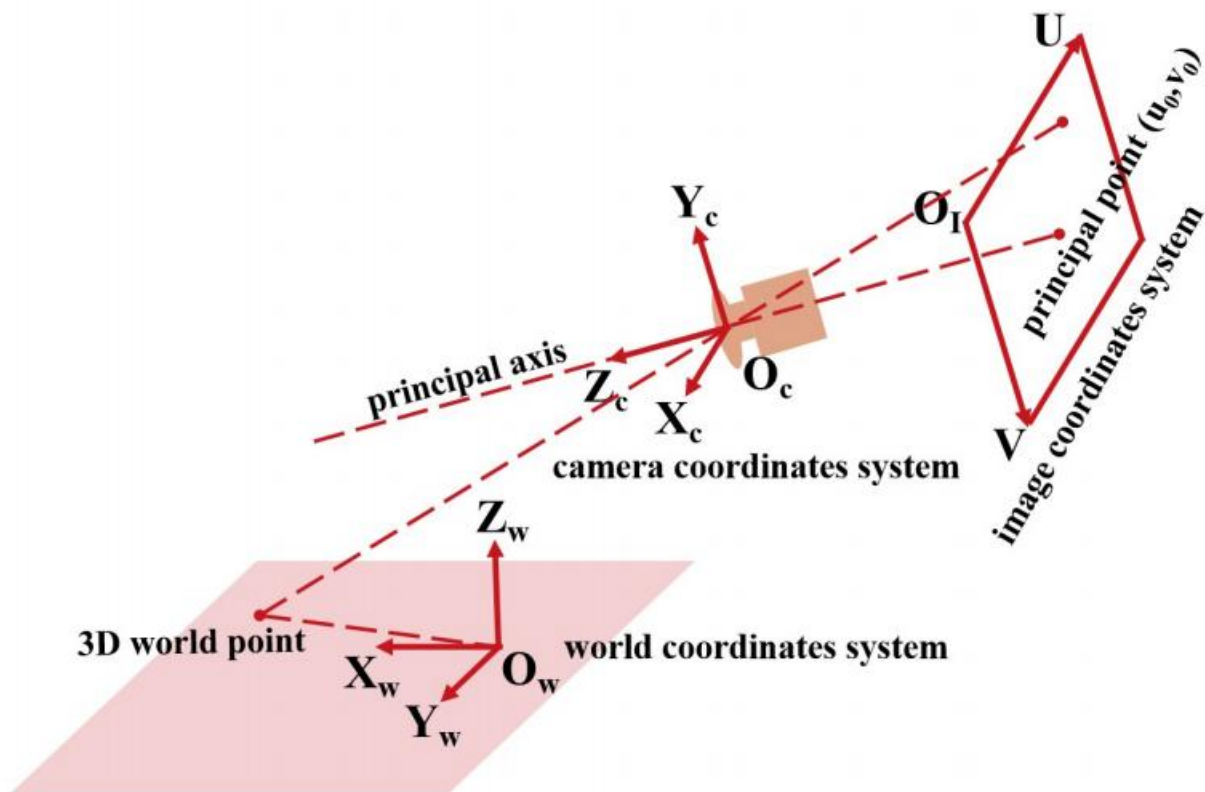
Revisit Common Sense of Projective Geometry



- *Invariance 1*: A point after perspective mapping (or perspective transform) is still a point.
- *Invariance 2*: A line after perspective mapping is still a line.
- *Invariance 3*: Coinciding lines after perspective mapping still coincide.
- *Invariance 4*: Colinear points after perspective mapping are still colinear.



Bijjective Mapping Between 2D Planes

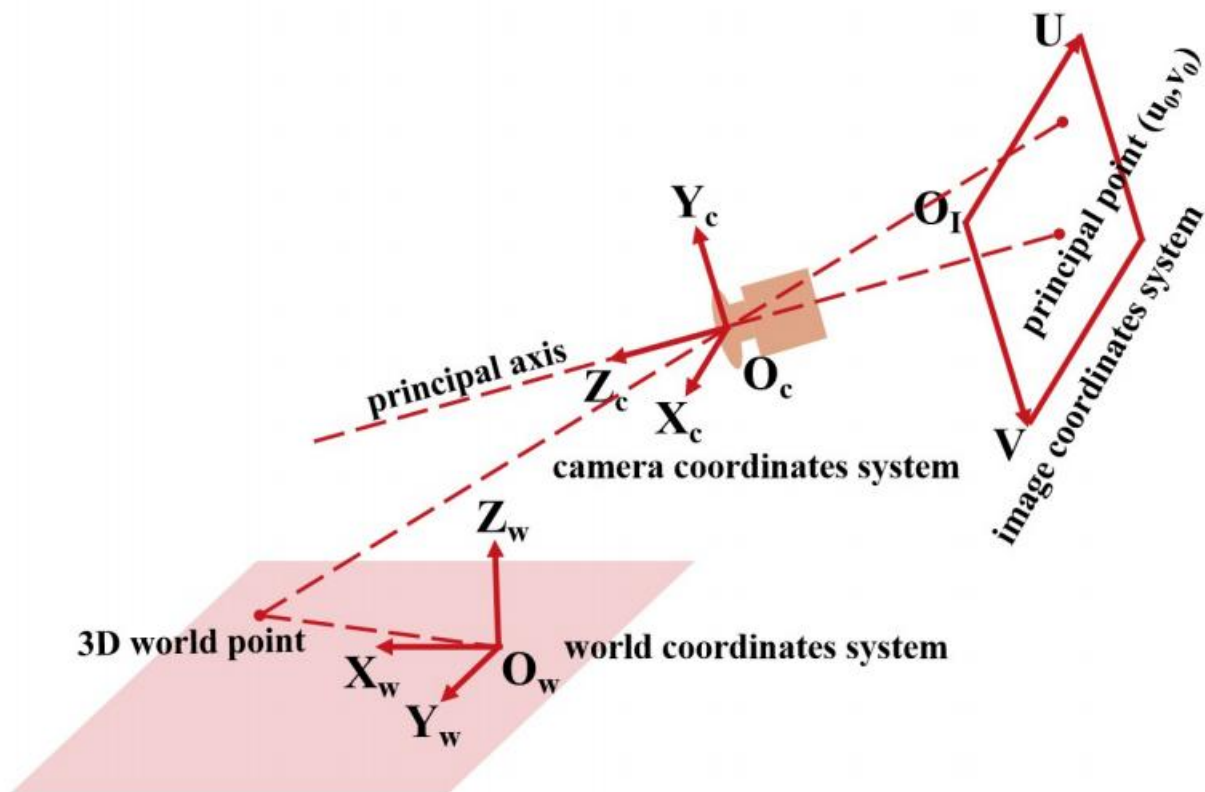


world coordinates system
(WCS)



image coordinates system
(ICS)

Bijjective Mapping Between 2D Planes



world coordinates system
(WCS)



image coordinates system
(ICS)

$$\lambda \mathbf{P}_i^A = \mathbf{H} \mathbf{P}_{wp}^A$$

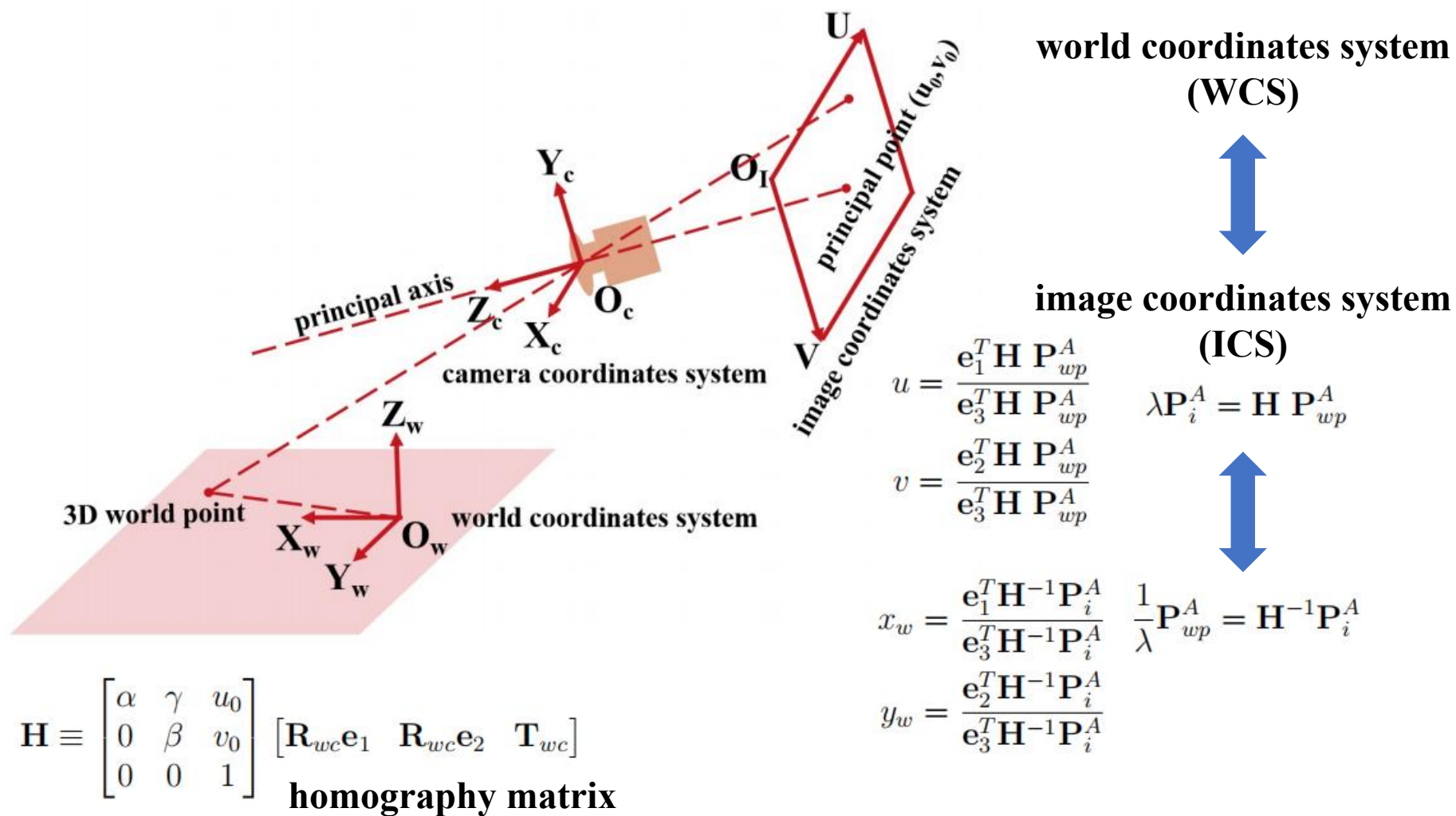


$$\frac{1}{\lambda} \mathbf{P}_{wp}^A = \mathbf{H}^{-1} \mathbf{P}_i^A$$

$$\mathbf{H} \equiv \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{R}_{wc} \mathbf{e}_1 \quad \mathbf{R}_{wc} \mathbf{e}_2 \quad \mathbf{T}_{wc}]$$

$$\frac{1}{\lambda} \begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} = [\mathbf{R}_{wc} \mathbf{e}_1 \quad \mathbf{R}_{wc} \mathbf{e}_2 \quad \mathbf{T}_{wc}]^{-1} \begin{bmatrix} \alpha & \gamma & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

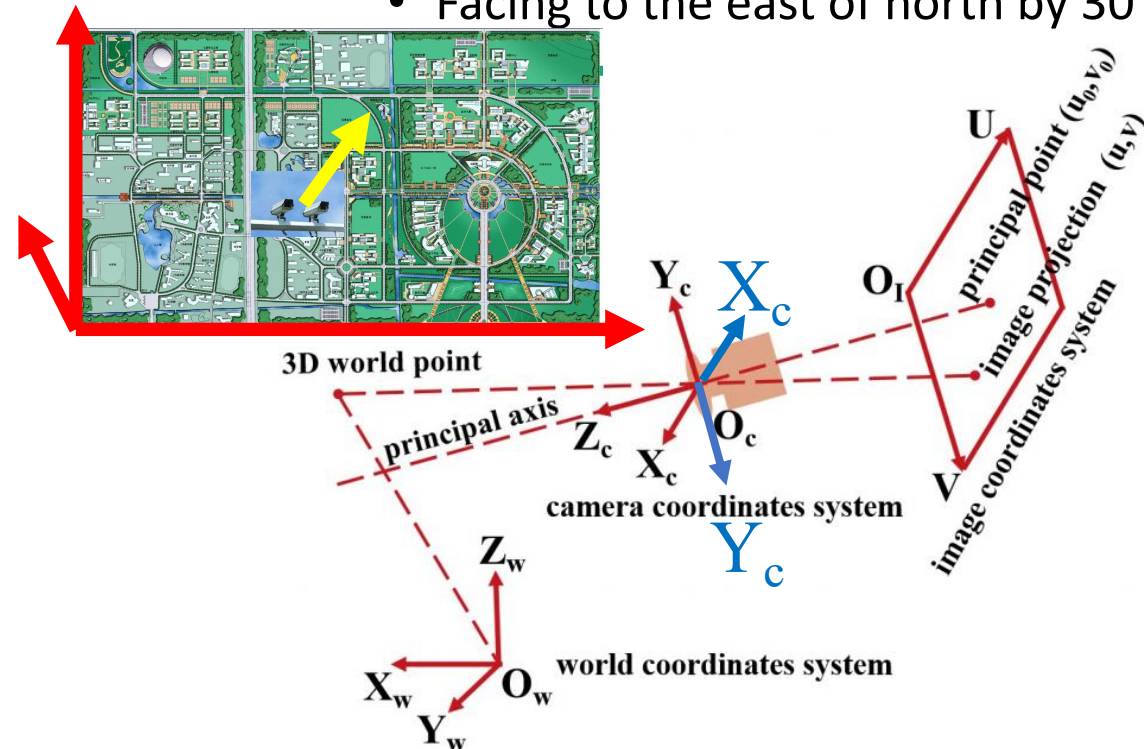
Homography



Comprehensive Example - Homography

- Street-View Camera

- “MapForUI.jpg” : 803x1404 pixels, each pixel representing 1.7 meters in the WCS
- Camera configuration
 - Located at (800, 600) meters, at a height of 10 meters
 - Facing to the east of north by 30 degrees, inclining downward by 15 degrees



- Positive Camera Intrinsic Matrix

- How should the CCS be pose?

$$\mathbf{A}_+ = \begin{bmatrix} -300 & 0 & 160 \\ 0 & -300 & 120 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{P}_{w1} = \begin{bmatrix} 825 \\ 643 \end{bmatrix} \quad \mathbf{P}_{w2} = \begin{bmatrix} 850 \\ 687 \end{bmatrix}$$

$$\mathbf{A}_+ = \begin{bmatrix} 300 & 0 & 160 \\ 0 & 300 & 120 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{P}_{i1} = \begin{bmatrix} 160 \\ 120 \end{bmatrix} \quad \mathbf{P}_{i2} = \begin{bmatrix} 200 \\ 80 \end{bmatrix}$$

DISCUSSION

Q & A